

June 27, 2023

CLOUD AND COMMUNICATIONS

Stock Rating:
OUTPERFORM

12-18 mo. Price Target \$35.00
EXTR - NASDAQ \$22.93

3-5 Yr. EPS Gr. Rate 17%
52-Wk Range \$25.13-\$8.58
Shares Outstanding 133.0M
Float 125.9M
Market Capitalization \$2,957.9M
Avg. Daily Trading Volume 1,748,122
Dividend/Div Yield NA/NM
Book Value NM
Fiscal Year Ends Jun
2023E ROE NM
LT Debt \$234.0M
Preferred \$0.0M
Common Equity \$103M
Convertible Available No

EPS Diluted	Q1	Q2	Q3	Q4	Year	Mult.
2022A	0.21	0.21	0.21	0.15	0.77	29.8x
2023E	0.20A	0.27A	0.29A	0.34	1.11	20.7x
2024E	--	--	--	--	1.53	15.0x
2025E	--	--	--	--	1.99	11.5x

Pro-forma EPS

Extreme Networks, Inc.

Major Beneficiary of Enterprise Edge Cloud and AI; Initiate at Outperform, \$35 PT

SUMMARY

We initiate coverage on EXTR with an Outperform rating and \$35 PT. EXTR provides critical edge infrastructure and software enabling the convergence of compute, network, and security, similar to Cloudflare's strategy in neutral datacenters. This is a nascent strategy but a much larger target market than its legacy networking, particularly for AI compute. Security will be the first major driver of growth, especially zero-trust the next year, followed by compute. Edge infrastructure is required for better performance/security, at lower cost with more enterprise control. In our view, EXTR is well positioned to capitalize on this trend as a provider of enterprise network/compute solutions from the edge to the data center, managed through its single cloud-based platform.

KEY POINTS

- **Cloud-first, Universal Fabric Platform.** EXTR's proprietary software and equipment are cloud-native and host-agnostic, delivering a 360-degree view of enterprise networks and its applications, users, and devices. EXTR was first in the industry to offer universal switching and access points, a substantially better solution than competitors'.
- **Cloud Continuum.** The modern enterprise post-COVID is highly distributed with an increasing number of users and devices at the edge. Applications and workloads that require low-latency should follow. EXTR is the first enterprise networking company to support a "cloud continuum" that can unify public, private, and edge cloud deployments. This allows data to reside locally on-prem vs. public cloud, resolving issues of data sovereignty.
- **Growing Market Share.** EXTR has identified a \$33B TAM that could double as compute and security converge on the same infrastructure. As an incoming challenger, EXTR is mainly targeting customers in the low-to-mid end market that are often overlooked by its primary legacy competitor CSCO. EXTR has only ~5% market share, but has been growing this 30bps+ per year.
- **Deep Partner Ecosystem.** The company's go-to-market is solely based on channel partners, which aligns with our 20-year horizontal segmentation investment thesis. EXTR has over 8K channel partners, and sells products that service its global footprint of 50K customers and 10M+ daily active users (DAUs).
- **Attractive Valuation.** We view EXTR as attractively valued at 15x 2024E P/E, a ~25% discount to the market, with the ability to grow revenue and EPS greater than 14% and 18%, respectively, over the next five years. Our \$35 PT is based on our DCF model, supported by a solid 7% FCF yield and strong balance sheet w/ backlog equivalent to over 2x quarterly revenue.

Stock Price Performance

Company Description

Extreme Networks provides end-to-end, cloud-native networking solutions. Extreme designs and manufactures network infrastructure equipment in conjunction with its proprietary cloud management platform and comprehensive suite of applications.

For analyst certification and important disclosures, see the Disclosure Appendix.

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5-YEAR PRICE PERFORMANCE



Source: Bloomberg

BASE CASE ASSUMPTION

- Revenue growth in mid-double digits, in line with long-term guidance
- Margin expansion of 100-200bps per year

UPSIDE SCENARIO

- AI and private network adoption accelerate demand for cloud-driven networking solutions
- Revenue growth outpaces 15% long-term guidance

PRICE TARGET CALCULATION

Our \$35 PT is based on applying a 17.6x multiple on our 2025E EPS estimate, supported by our DCF model which assumes an 11.1% WACC, 2% long-term growth rate, and 2034E terminal EBITDA value of \$814M.

KEY RISKS TO PRICE TARGET

Key risks to our price target include: 1) highly competitive industry underscored by rapid technological innovation and consolidation; 2) M&A integration; and 3) hardware component shortages.

INVESTMENT THESIS

Extreme has one of the best Wi-Fi and Switching equipment and software for enterprises. This is provisioned and managed over the cloud. Its new universal equipment supports compute, security, and now AI, all at the edge and directly in the hands of enterprises. This is now a substantially larger TAM, and in our view, EXTR has the ability to gain market share.

CATALYSTS

- Once in a decade enterprise network upgrade cycle driven by new wireless technologies and spectrum
- Large increase in available wireless spectrum
- Major logo wins

DOWNSIDE SCENARIO

- Prolonged component shortages and supply chain issues
- Industry consolidation favors larger service providers with more scale and resources

ESG CONSIDERATIONS *

ESG Rating	C+
ESG Rating Environment	C+
ESG Rating Social	C
ESG Rating Governance	B

*Environmental/Social/Governance (ESG) scores are courtesy of Refinitiv's ESG product which are designed to transparently and objectively measure a company's relative ESG performance, commitment and effectiveness, based on company-reported data. Refinitiv's ESG ratings are independent of Oppenheimer's stock ratings and are not taken into consideration when assigning a rating. For full details about Refinitiv's ESG scores, please [click here](#)

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Investment Thesis

Extreme Networks is an enterprise network provider that is entering compute, security and now AI—all provisioned at the enterprise location, but managed over the cloud. Its unique cloud-native services simplify and consolidate private network builds and management. This can be seen in the company motto of “one network, one Cloud, one Extreme.” We believe Extreme Networks, as a neutral cloud provider, is in a virtuous cycle which provides the company with a competitive advantage. We have a constructive view on Extreme Networks based on:

- Purely enterprise-focused with most flexibility and innovation;
- Full range of managed services from the datacenter to customers premise over Extreme Fabric;
- Premium and integrated wireless and wireline products (switches, routers, and wireless access points), managed over one universal cloud platform (Extreme Fabric);
- Strong demand for high quality enterprise wireless networks with greatly improved spectrum and 5G/WiFi technologies;
- Private wireless networks are seeing strong demand due to mission-critical applications needing dedicated bandwidth than can be designed to operate around physical barriers;
- Pure play on migration of Cloud/AI to the Edge. Its Cloud services are five years old, which enables the company to innovate rapidly, and to leverage its infrastructure with new services—edge compute in particular is required to manage robotics, IOT, AI and security;
- Cloud-agnostic and hybrid cloud, its compute infrastructure can support any hyperscaler or SaaS application;
- It is more focused and nimble than its competitors, and as a result is the least expensive and faster provider of this new triple play for enterprises;
- Strong partnerships on cellular, security, and chips (Broadcom’s new Tomahawk 51TB chips with embedded photonics);
- Room to grow market share from current ~5% through superior equipment and services.

We expect the company to benefit from favorable near-term catalysts:

1. An aggressive migration of networking to the cloud and now edge, much as we have seen for compute;
2. Edge should enable EXTR to sell more compute and security services;
3. Easing supply chain bottlenecks;
4. A large increase in wireless spectrum (both 6 GHz WiFi and 5G/LTE CBRS spectrum), and new wireless standards (6E & 7);
5. Potential to enter the 5G equipment business; the company currently partners with Ericsson and others while it provisions the WiFi and switching;
6. Customers looking for easy-to-use complete network management tools integrated with AI to improve performance and security;
7. More flexible partnership and licensing and API strategy than peers’—it charges one license per device that covers all its cloud services; and
8. Operating leverage from lower equipment transport costs, price increases, new services, and a flexible unified cloud-based platform.

Long-term, on-premise enterprise networks are the most critical input into IT, and are seeing a step function improvement in capacity and latency. This is enabling last mile innovation, converging cloud and communications services, and more

specifically, converging networking, security and edge computing onto one cloud-controlled infrastructure. This improvement will drive a whole set of new automated use cases leveraging AI. AI inferencing is more efficient to do as close to the customers as possible to run robots and dozens of new applications. EXTR's wireless hotspots can be powered and managed over the cloud enabling customers to manage access, security, applications remotely and efficiently. The company has clear visibility on how each device and application on the device is performing. It can also control data completely and is integrated with some of the best switching capability in the industry, particularly for high definition video.

The company has guided to 15% annual revenue growth and 200 basis points margin expansion in the next few years, but we see this as conservative given its \$500M-plus backlog (vs. \$100M historical), normalized volume growth in the 5-10% range, and ARPU/pricing increases of 5-10% as well as:

- We expect private enterprise networks to grow in the 5% range long term given the productivity improvements enabled by Cloud/AI. With edge compute and security, we think the market growth is more like 10%, and EXTR will likely gain share;
- Enterprise network hardware represents less than 2% of global IT spend, yet is a critical input;
- Network upgrades are needed to support new use cases over the cloud-AI, Robotics, IOT;
- Extreme has been and will likely continue to gain share given its premium products, services, simplified licensing, better support and lower prices than competitor Cisco; and
- Positive secular tailwinds that give EXTR pricing power. The company has already raised prices by ~10% for equipment last October (which accounts for 70% of its revenue), but not yet for services (which accounts for 30% of its revenue).
- The most important drivers though will be adding new service revenue from security, AI and compute.

Cloud technologies are transforming businesses across all industry verticals, impacting communication, productivity, operations, and more (see our past reports [here](#), [here](#), [here](#) and [here](#)). In fact, using cloud for enterprise networks was the only way to support a distributed workforce during the pandemic. In public venues, consumers will demand high quality connectivity or favor places that have it. Extreme is using AI to improve its own service (predictive failures more customized application acceleration) and offer new security features which at their base are really forms of AI/analytics. It also helps enterprises collect more data over their networks to improve productivity and drive new revenues.

Private enterprises are looking to combine the benefits of increased automation and efficiency from the cloud with the security and low-latency of a 5G network, which will be complementary to WiFi networks. Enterprises want to control their networks and data and integrate them with IT solutions such as SD-WAN and IP applications. All this places Extreme Networks in a very favorable position, in our opinion. We think the stock is attractively valued at approximately 15x 2024E calendar earnings, with one of the strongest balance sheets in our coverage universe (essentially no net debt), and \$250M in estimated FCF in 2025, a plump 7% FCF yield growing 20% per year.

The company's cloud platform is called Extreme Fabric, and is designed to enable innovation and flexibility. It is a unified wired/wireless/SD-WAN platform that can integrate any third-party device or cloud. It gives seamless control from the core right to the Wi-Fi access point and devices. The company was first to

market with this unified platform strategy for enterprises, and has been first with many other services like digital twin. This platform also makes it easy to perform AI, integrate any security service, manage IoT, and extend to branch office locations.

Investment Positives

1. Enterprise networking is a large market that could potentially double or triple in size with compute and security added on to the same infrastructure.
2. Extreme has small market share, but is growing this by ~30bps+ per year.
3. Software and Cloud are less than 20% of revenues, but are growing 2x-3x faster than the products segment; AI should support this growth with much higher margins.
4. Long-term demand and spend on Cloud will likely be strong and resilient, and edge cloud where EXTR plays should grow faster. We believe Cloud represents the third general-purpose technology in 300 years and will disrupt virtually every industry vertical.
5. Extreme is an established leader in innovation, holding over 500 patents.
6. Extreme has an excellent acquisitions track record, which we think will be critical going forward given the cloud/networking/security convergence.
7. Extreme should benefit from a once-a-generation upgrade in enterprise networks, helped by new wireless spectrum and wireless technologies, all leveraging the cloud including 5G, IoT, and WiFi 6, 6E, and 7, all run over one flexible cloud platform. This upgrade cycle gives it strong pricing power.
8. The supply chain has opened up, and shipments and revenue should grow well above trend. The company has \$500M in backlog tailwind that should enable it to beat its LT revenue and margin guidance the next three years.
9. Margins are expanding given lower transport costs, higher pricing, and more software-based revenues.
10. The IoT market is expected to grow quickly at 16% '20-'25E CAGR to over \$820B global TAM. IoT can leverage EXTR's infrastructure with high operating leverage.
11. AI inferencing models will be primarily done at the edge, which will require secure, low-latency connections. We may also see a lot of AI training at the edge in a reverse CDN-type architecture.
12. The company serves relatively stable industries, with 65% of revenues from government and enterprise and an additional 15% from very stable industries (sports, healthcare, etc.).
13. Valuation appears relatively attractive at ~15x 2024E EPS and 10x 2026E.

Investment Concerns

1. The convergence of cloud and communications is creating risks of a major revamp of services and potentially new competitors.
2. The communications industry is undergoing considerable consolidation, with most of the market being served by a limited number of cloud service providers (CSPs) often with more resources and scale.
3. Extreme is subject to supply chain issues, particularly to prolonged semiconductor chip shortages. The company has navigated industry-wide constraints, but sources select components from a limited number of suppliers and lacks long-term agreements with fixed pricing.
4. Economic slowdown could slow network upgrades.
5. The nature of Extreme's industry is one of intense competition driven by rapid technological changes.
6. Roughly 50% of Extreme's revenues are derived from outside of the US, posing foreign currency and regulatory risks.
7. Cisco, its main competitor, could look to bundle services together more aggressively.
8. Telco wireless providers would like to enter the private wireless/network market using 5G to enable new use cases (automated factories, etc.).
9. It is not entirely clear the new use cases and monetization opportunities that require significantly higher network and lower latency yet and or where AI inferencing will occur.
10. Shift to Network as a Service (NaaS) is an opportunity, but also a risk. New entrants in the enterprise network market are aiming to disrupt legacy operators with new services designed specifically for NaaS.

Company Description

Founded in 1996, Extreme Networks, Inc. (EXTR) is a leading provider of cloud-based, on-premise, and hybrid networking, and increasingly, security and compute services. It is headquartered in Morrisville, NC. The company designs, develops, manufactures, and operates wireline and wireless network infrastructure equipment and software, and increasingly manages this equipment remotely using its proprietary cloud software applications. These applications leverage AI to deliver network policy, security, and access controls. Extreme Networks has made over a half dozen strategic acquisitions to create these capabilities, and we believe more may be necessary to maintain its competitive edge.

The COVID-19 pandemic accelerated the adoption of new IT delivery models and applications by organizations across nearly every industry vertical. As a result, networks have become more critical than ever to operations and strategy, as organizations process more data from more sources, connect to more devices, and run more SaaS applications. In response to these evolving needs and challenges faced by customers, EXTR has placed particular emphasis on

simplifying the migration process to hybrid-cloud IT models. The new wave of "Infinite Enterprises" identified by EXTR requires a flexible, secure, scalable, and customizable network that can be accessed anywhere at any time, necessitating fundamental modifications and enhancements to provide real-time connectivity to distributed workforces. EXTR's technology stack enables enterprises to embrace cloud and digital transformation without replacing existing infrastructure, which can be costly and resource-intensive.

Exhibit 1. Extreme Networks Historical Timeline

Year/Date	Event
1996	Founded in Cupertino, CA
Oct. 1996	Acquired Mammoth Technology
Apr. 1999	IPO listed on NASDAQ stock exchange under EXTR ticker for \$17 per share
2001	Acquired Optranet and Webstacks for \$73M and \$74M, respectively
2002	Announced its first 10 GB Ethernet product
2013	Acquired Enterasys Networks for \$180M
Nov. 2014	Named first official Wi-Fi solutions provider of the NFL
Apr. 2015	Appointed current and active CEO, Ed Meyer cord
Oct. 2016	Acquired Zebra Technologies' Wireless LAN business for \$55M
2017	Acquired Avaya's networking business in transaction valued at \$100M
2017	Acquired Brocade's SRA (Switching, Routing, and Analytics) business from Broadcom
2019	Acquired Aerohive Networks for \$272M
Oct. 2019	Announced its ExtremeCloud IQ cloud-based enterprise network management application
Oct. 2019	Selected Broadcom's networking chips and programmable silicon to power its extensive portfolio of hardware products
Jun. 2020	ExtremeCloud IQ became the only 4th-generation cloud platform across all major CSPs (including AWS, Azure, and GCP) with an unlimited data offering
Sept. 2020	Announced universal platforms for enterprise-class switching and wireless infrastructure
Feb. 2021	Selected by MLB as official Wi-Fi solutions provider
Jul. 2021	Became first vendor in industry to ship an enterprise-grade Wi-Fi 6E solution
Sept. 2021	Acquired Infovista's Ipanema SD-WAN business
Nov. 2021	Joined NASCAR's Technology Partner Platform
Jan. 2022	Announced multi-year partnership with the NHL as official Wi-Fi solutions provider
May 2022	Partnered with Verizon Business to deliver connectivity and network insight solutions for large venues and stadiums across EMEA
Jan. 2023	Named leader in Gartner's Magic Quadrant for Enterprise Wired and Wireless LAN Infrastructure for fifth-consecutive year

Source: Oppenheimer & Co. Inc., Company Reports

As mentioned, EXTR has a long history of successfully acquiring and seamlessly integrating new assets into its product portfolio, dating back to before its IPO in 1999. The company also has an extensive network of partners through its ExtremeAlliance ecosystem focused on driving partner-built integrations and solutions to support customers on their path to becoming an "Infinite Enterprise."

Exhibit 2. Extreme's Targeted Verticals

LEADERS IN TARGETED VERTICALS RELY ON EXTREME



Enterprise Vertical	Contribution*	Key Customers
Government / Education	>40%	
Enterprise and Other	~25%	
Manufacturing	~10%	
Healthcare	~10%	
Retail and T&L	~10%	
Sports and Entertainment, Hospitality	~5%	

*Contribution Based on Q2 FY23 Bookings

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Source: Company website

EXTR's global footprint services 50K customers and 10M-plus daily active end users across a variety of industry verticals, primarily targeting: Government & Education, Enterprise, Manufacturing, Healthcare, Retail, Transport & Logistics, and Sports, Entertainment, and Hospitality. Over 50% of EXTR's revenue comes from Government & Education and Healthcare deployments where healthy demand for wireless and campus switching upgrades continue to serve as durable tailwinds. Though only representing 5% of revenues, EXTR is taking share as it deepens its relationships with customers in Sports and Entertainment, notably announcing several partnership agreements with top professional leagues including the MLB, NFL, NHL, and NASCAR, plus dozens of Division I NCAA universities.

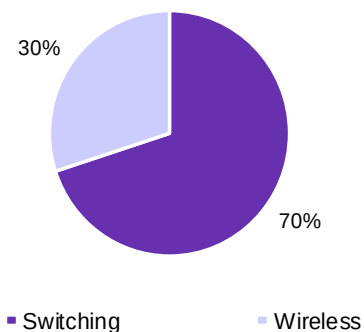
EXTR's go-to-market strategy is solely focused on sales channel partners, with no direct sales from an internal salesforce. This approach aligns perfectly with our 20-year horizontal segmentation preference for investing. Currently, the company has over 8,000 partners selling its products, with only approximately 20% of these partners offering the full suite of EXTR's products. However, the company is actively seeking to increase this number. As a result of its channel go-to-market strategy, EXTR is able to optimize all its operating support and billing systems around this one channel, with no channel conflict. This approach also enables the company's channel partners to bundle best-of-breed services from other suppliers.

As an up-and-coming challenger in the networking space, EXTR is focusing on capitalizing on opportunities in the mid- and low-end segments of the market, which larger incumbent competitors tend to overlook. Despite its ~5% market share and sub-2% revenue share, the company is experiencing double-digit revenue growth, strong forward-looking bookings, and a product backlog equivalent to 2.5 times quarterly revenue that is in the early stages of converting to revenue as product lead times continue to improve. While the product book-to-bill ratio had been above 1 for the past two years, it has now dropped below 1, indicating that the company is executing well on converting its backlog into revenue.

Product and Services Overview

EXTR offers an end-to-end portfolio (Exhibit 4) of cloud-driven enterprise network solutions from the edge to the data center. EXTR designs and manufactures proprietary networking equipment including routers, WLAN (Wireless LAN) controllers, wireless access points (APs), and wired/wireless switches on top of an extensive repertoire of AI-enhanced software applications for access control, network analytics, and network management that can be easily monitored and controlled on virtually any device over the cloud. Digital twin capabilities, which enable enterprises to virtually simulate physical models in a virtual environment, are unique to EXTR in the networking vendor space. Also unique to EXTR is the option for customers to choose on-premise or cloud deployments—most competitors, such as Cisco Meraki and Juniper Mist are cloud-only. The combination of EXTR's in-house hardware, powerful software, multivendor integration, and deep partnerships support enhanced performance, security, and agility demanded by end users across connected organizations.

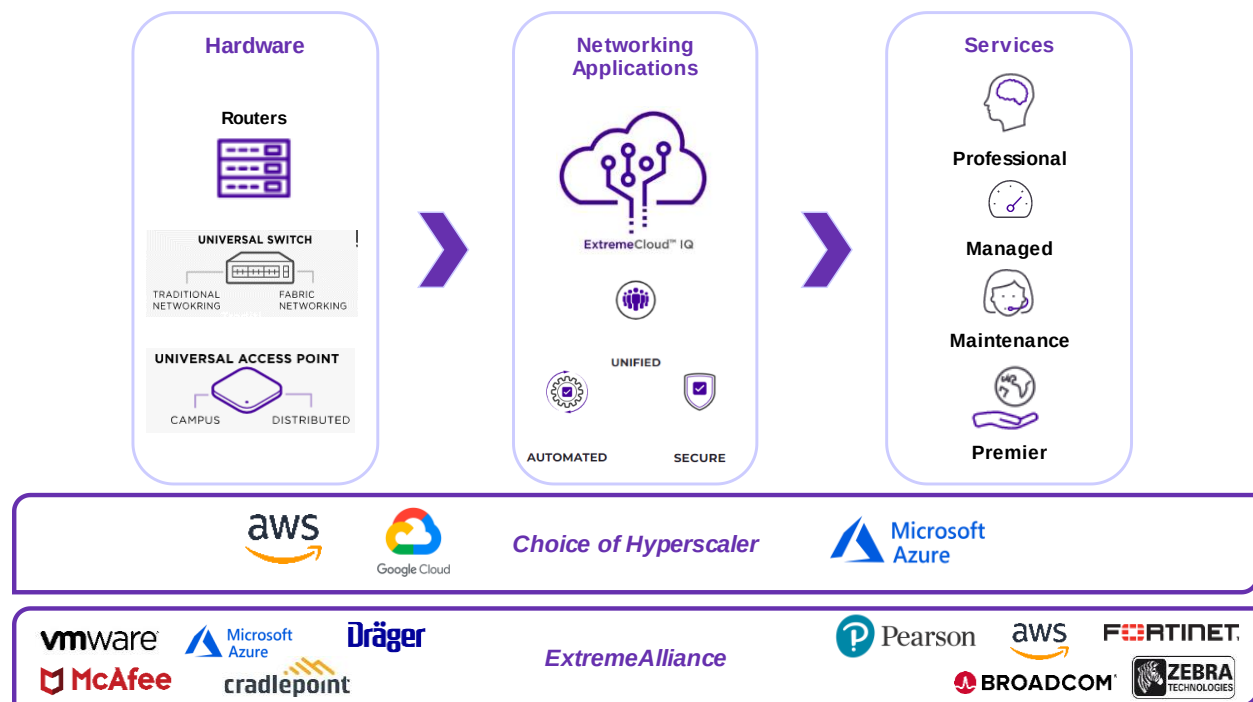
Exhibit 3. Estimated Revenue Breakdown by Product



Source: Oppenheimer & Co. Inc. estimates

EXTR's Fabric Connect technology equips enterprises with the tools to reduce lead times for completing network deployments, quickly identify and resolve issues in the network or application layer, and minimize network administrator workloads through ubiquitous automation. It provides a unified management plane for Wi-Fi and switching, with plans to extend to SD-WAN. EXTR has successfully deployed thousands of fabric networks globally that are industry-aligned and standardized, capable of serving all market sizes across all market verticals.

Exhibit 4. Extreme's End-to-End Solution Stack



Source: Company website, Oppenheimer & Co. Inc.

ExtremeCloud IQ is the company's cloud-native, software-based, network management and intelligence platform. Designed to streamline every aspect of the network from deployment to maintenance, ExtremeCloud IQ delivers a 360-degree view of the network and its applications, users, and devices using real-time and historical monitoring, network troubleshooting, prediction, and AI/ML insights generated from millions of network and end user data points in addition to supporting open APIs for developers to create third-party applications. It is cloud-hosting agnostic which entails support for AWS, Microsoft Azure, and GCP. Moreover, Extreme offers consistent features and capabilities across public cloud, private cloud, and on-premise deployment types, giving customers flexibility as their needs and strategies change. Enterprises that want to integrate AI in their networks, particularly for IoT applications where security is a top concern, will require a cloud-based network with powerful management software such as ExtremeCloud IQ.

However, many organizations around the world still rely on legacy networks based on protocols originating ~30 years ago that were designed to support simple, static workflows. In these instances, the network can become a bottleneck for modern enterprises that are increasingly dynamic and distributed. EXTR's network fabric product address these legacy issues and help transform networks from pain points to powerful assets that drive growth and productivity. Fabric networking solutions are highly adaptable, virtualized, and scalable to meet the complex and ever-changing networking needs of today's enterprises. Network fabrics consist of the physical underlay made up of switches, routers, and connected devices with a virtualized overlay used to transport data. A few universal benefits of fabric networking include: 1) improved time to service via network automation; 2) enhanced security from network segmentation; 3)

improved resilience from a reduced number for network protocols; 4) more consistent quality through a unified campus network that seamlessly integrates wired and wireless networks; and 5) end-to-end visibility and data collection.

Exhibit 5. ExtremeCloud IQ Subscription Tiers and Features

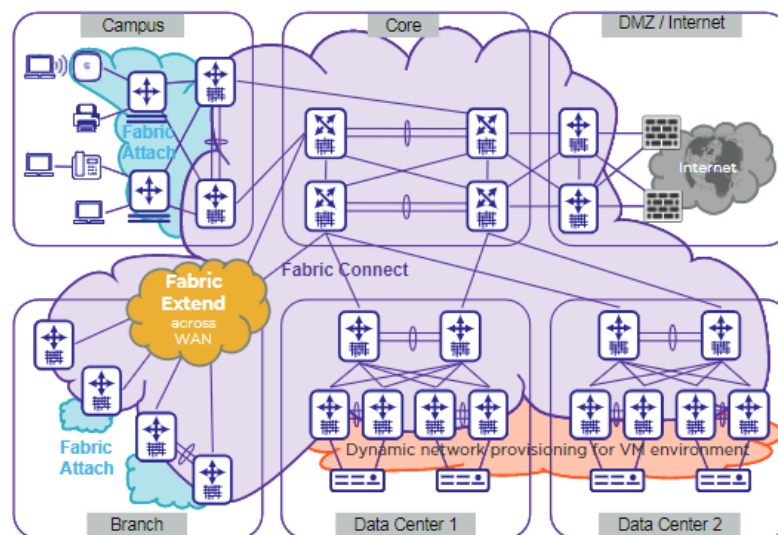
	ExtremeCloud™ IQ Connect	ExtremeCloud™ IQ Navigator	ExtremeCloud™ IQ Pilot*	ExtremeCloud™ IQ CoPilot*
Availability	 PUBLIC	 PUBLIC	   PUBLIC PRIVATE LOCAL	 PUBLIC
Feature Set	Free device management for access points, switches, and routers	EVERYTHING IN CONNECT, PLUS: Additional management functions for third party and non-native cloud devices	EVERYTHING IN NAVIGATOR, PLUS: Advanced infrastructure management, reporting, and remediation tools	EVERYTHING IN PILOT, PLUS: Explainable machine learning derived insights and intelligence. Algorithmically detected anomalies:
Unique Features	• Device onboarding • Guided Configuration • Centralized Management • Wi-Fi Planner • Basic Monitoring Tools • Essential Security	• Application Visibility • SSH Proxy • Web Proxy to WING Controller • Web Proxy to Extreme Campus Controller	• Advanced Topology View • Troubleshooting Heuristics • Contextualized Optimization • Role Based Profiling • Advanced 360° Reporting • Comparative Analytics *only available for cloud native devices	• Digital Twin • Wired/Wireless AIOps • Anomaly Detection • Explainable ML • Remediation Recommendations • Automated GTAC Support • Client Experience *only available for cloud native devices

Source: Company website

ExtremeCloud Edge

Building on its “one network, one cloud” philosophy, EXTR has expanded its cloud management platform with its new ExtremeCloud Edge service announced in May at the company’s Connect Conference. ExtremeCloud Edge is the industry’s first networking cloud continuum that delivers flexible, unified management capabilities that spans across public, private, and edge cloud deployments. Enterprises have become significantly more distributed with an increasing volume of users and devices at the edge that rely on bandwidth-intensive and latency-sensitive applications that will require the seamless migration of networking applications to the edge. ExtremeCloud Edge unifies EXTR’s core portfolio of applications and tools including ExtremeCloud IQ, CoPilot, and SD-WAN. ExtremeCloud Edge resolves cloud/data sovereignty concerns, allowing data to reside local and on-prem versus the public cloud. The company indicated ExtremeCloud Edge will be available beginning summer 2023 for select partners before becoming generally available in early 2024.

Exhibit 6. Extreme Fabric Connect Infrastructure

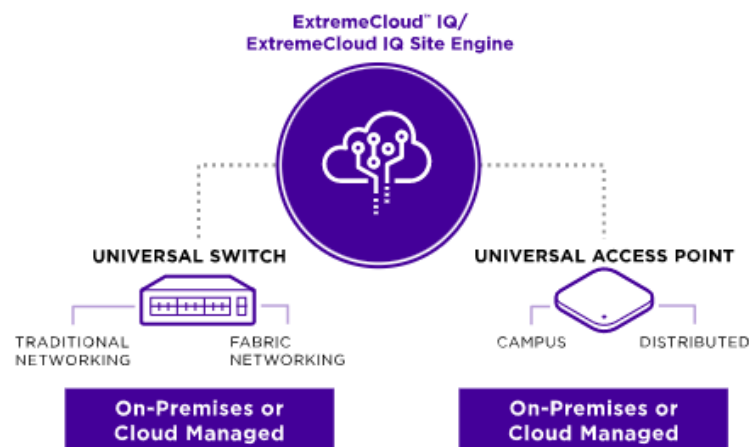


Source: Company website

Universal Platforms for Switching and Wireless

In September 2020, EXTR became the first in the networking industry to offer universal platforms for its enterprise class switching and wireless portfolios. High performance, high efficiency access points and switches deliver secure end-to-end and edge-to-core operations that can be fully managed and optimized via the ExtremeCloud IQ management software. These innovative products greatly simplify the ordering and licensing process for customers. As business needs change, enterprises may need to change operating systems which typically entails costly rip and replace. Universal platforms eliminate this step through backwards compatibility and can run multiple operating systems allowing EXTR to deliver the same functionality while reducing TCO for the customer.

Exhibit 7. EXTR Universal Switching and Wireless Platforms



Source: Company website

EXTR's many acquisitions have forced it to mix a lot of functionality from install bases to its core platform. Third-party migration capabilities is a competency many of its competitors struggle with, namely Cisco whose Meraki cloud management platform is not fully integrated with all Cisco Catalyst products.

The consolidation of all disparate product lines EXTR has acquired over time has significantly helped the company navigate supply chain shortages. Fewer stock keeping units (SKUs) results in fewer components to chase, key to alleviating backlog. Today, about 60% of SKUs are universal which EXTR anticipate will climb to 90% by year-end, while 40% of actual bookings being shipped are universal.

Wi-Fi 6E and 7 Drives Network Upgrades and New Apps

Wi-Fi is a deeply entrenched technology that has had a universal impact on virtually every consumer and enterprise. The market for enterprise wireless LAN infrastructure equipment alone is a \$10B TAM, historically growing in the mid-single digits and is supported by mature management platforms and is familiar to IT professionals. Each generation delivers new enhancements and improvements to speed, coverage, spectral efficiency, and power consumption.

Wi-Fi 6E is the latest generation of Wi-Fi and is a major upgrade from 5 and even 6. The most significant innovation is the ability to connect to the 6 GHz band; it also provides more bandwidth, supports new mesh use cases, is more reliable and secure, and can support more devices per access point than Wi-Fi 5. 6 GHz is a clean spectrum band, an exclusive channel dedicated to 6E devices—a first for unlicensed Wi-Fi use. Wi-Fi 6E will be more effective for dense urban deployments in which interference from traditional radio frequencies and traffic from legacy devices causes congestion and disruptions. However, 6E is not standards-based due to the fact that the 6 GHz spectrum will not be addressed until 2024 when 802.11be (Wi-Fi 7) is ratified.

Wi-Fi 6E is garnering massive interest among enterprises globally. The main driver behind this is the lack of available spectrum allocated to cellular in addition to licensed spectrum buildouts being expensive. About 14M enterprise grade access points are shipped per year while about 80M have been installed. Demand for wireless APs is strong and will likely continue during this next major upgrade cycle to 6E and 7. Advances in Wi-Fi standards, increased spectrum, and advancements in equipment are blurring the lines between Wi-Fi and wireless; however, we expect the two technologies to continue to converge and coexist, with enterprises deploying private networks leveraging both 5G and Wi-Fi.

In July 2021, EXTR was the first in the industry to ship an enterprise grade Wi-Fi 6E solution. More recently, EXTR partnered with Comcast to deploy a Wi-Fi 6E ready network at Oracle Park, making it the first professional sports venue equipped with the technology. EXTR is looking to deploy about 900 Wi-Fi 6E access points throughout the ballpark which will support enhanced fan experiences. Both instances demonstrate EXTR's ability to outpace competitors in innovation underpinned by its cloud-first and universal platform strategy.

Exhibit 8. Wi-Fi 6E Delivers Significant Advantages over Wi-Fi 6

	Wi-Fi 6	Wi-Fi 6E	Wi-Fi 6E Advantage
Key Features	Orthogonal frequency-division multiple access (OFDMA) Multi-User Multiple-Input Multiple-Output (MU-MIMO) 1024 QAM	All features of Wi-Fi 6 included	All Wi-Fi 6 features that improve performance, speed, coverage, efficiency, & power savings are included in Wi-Fi 6E and further enhanced due to the ability to operate in 6 GHz band
Frequency Bands	2.4 GHz 5.0 GHz	2.4 GHz 5.0 GHz 6.0 GHz	• More reliable critical application performance with less interference in 6 GHz, even in dense environments • Enhanced capability for mesh applications with clean 6 GHz spectrum • Legacy device performance can improve, as more new devices offload from 2.4 & 5 GHz and transition to 6 GHz
Security Standards	WPA3 (Optional) Enhanced Open (Optional)	WPA3 (Mandatory) Enhanced Open (Mandatory)	• Better IoT security • Improved security for guest devices
Maximum 20 MHz channels available	28 channels (across 2.4 and 5 GHz)	87 channels (across 2.4, 5, and 6 GHz)	• More capacity for high-bandwidth applications • Spectrum power rules conducive for the use of 80 MHz and 160 channels in the enterprise • Better use experiences and performance for emerging applications such as HD video conferencing, VoIP and VR/AR

Source: Extreme Networks

Wi-Fi usually follows a five- to six-year upgrade cycle (Exhibit 9) and entails upgrading the supporting wired and switching infrastructure, which also averages around a six-year upgrade cycle. Wi-Fi 7 (802.11be) is on the horizon with standards anticipated to be adopted in 2024, though Huawei has already begun shipping Wi-Fi 7 equipment this year, but likely won't be meaningful until at least next year, especially when compatible smartphones will not become available beginning at some point in 2H24. It is speculated that Wi-Fi 7 will ultimately deliver connection speeds of up to 46 Gbps that will be able to support emerging use cases such as ultra-low latency extended reality (XR) applications, cloud-based gaming, multicast video streaming, and 8K resolution video. Aside from superior speeds, capacity, and latency, Wi-Fi 7 will support 320 MHz channels, multi-link implementation, and 4096 QAM. It is possible many enterprises will wait for Wi-Fi 7. The install base so far for 6 GHz is small as it takes time for devices to catch up. An estimated 10-15% of Wi-Fi equipment shipments in 4Q22 were 6E, and EXTR expects it can soon account for 25% of its sales volumes.

Gartner estimates that about 30% of enterprises have built out networks using Wi-Fi (95% of which are Wi-Fi 5), while two-thirds of new network builds are Wi-Fi.

Currently, only ~15% of incremental shipments are for 6/6E equipment, implying ample runway for EXTR to take share as an early leader and first mover in the space, especially as enterprises move away from wired connections. Historically, companies tended to restrict Wi-Fi to a limited space, but now are deploying across entire office spaces and campuses which will further drive demand.

Exhibit 9. Wi-Fi Undergoes Upgrade Cycle about Every ~5 Years

Wi-Fi generations				
Generation	IEEE standard	Adopted	Maximum link rate (Mbit/s)	Radio frequency (GHz)
Wi-Fi 7	802.11be	(2024)	1376 to 46120	2.4/5/6
Wi-Fi 6E	802.11ax	2020	574 to 9608 ^[41]	6 ^[42]
Wi-Fi 6		2019		2.4/5
Wi-Fi 5	802.11ac	2014	433 to 6933	5 ^[43]
Wi-Fi 4	802.11n	2008	72 to 600	2.4/5
(Wi-Fi 3)*	802.11g	2003	6 to 54	2.4
(Wi-Fi 2)*	802.11a	1999	6 to 54	5
(Wi-Fi 1)*	802.11b	1999	1 to 11	2.4
(Wi-Fi 0)*	802.11	1997	1 to 2	2.4
*(Wi-Fi 0, 1, 2, 3, are unbranded common usage) ^{[44][45][46][47]}				

Source: Wikipedia

Private Enterprise Networks

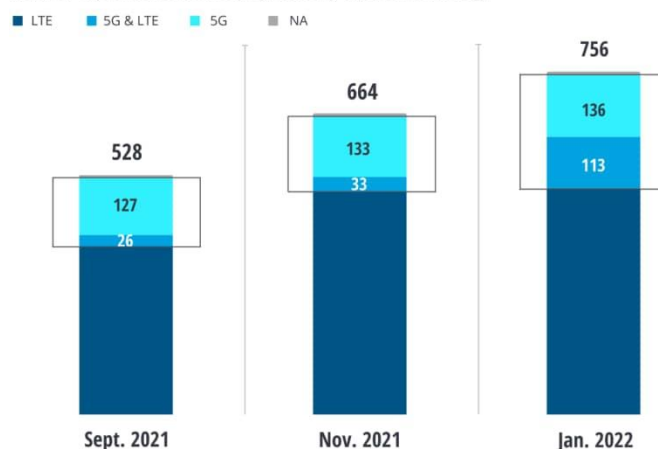
Enterprise networks refer to IT infrastructure that connects users, devices, data, and applications for campus deployments. These are purpose-built networks that enable organizations to control and manage their networks and data, connect mobile and fixed users/devices, and deliver custom fit solutions for targeted business outcomes bolstered by enhanced security, transparency, and latency. The carriers are in the early stages of aggressively attacking the business market as management teams within the industry have shown interest in these networks from enterprises is strong and growing attributable to ongoing digital transformation programs, despite rollouts proceeding slower than expected. According to a survey from Deloitte (Exhibit 10), there were over 750 private network deployments as of January 2022, with an increasing mix of these networks beginning to experiment with a combination of LTE and 5G. We expect this trend to continue and accelerate with more networks depending on 5G and/or Wi-Fi 6 and 6E. These wireless technologies will unlock new use cases not possible on LTE and previous generations of Wi-Fi. Today, the primary application for private 5G networks are for IoT connections in the automotive, mining, sea port, and manufacturing industries, including automated robotics and equipment sensors that collect data used to optimize system processes.

The key question will be if enterprises choose the carriers for their deployments or if they would rather own and control their own spectrum and equipment.

Today, less than 25% of enterprises rank mobile network operators as their preferred partner for building a private network, according to a survey from the GSMA, a major opportunity for players like EXTR.

Exhibit 10. Interest in Private Enterprise Networks Is Rising

Number of private network deployments by wireless technology

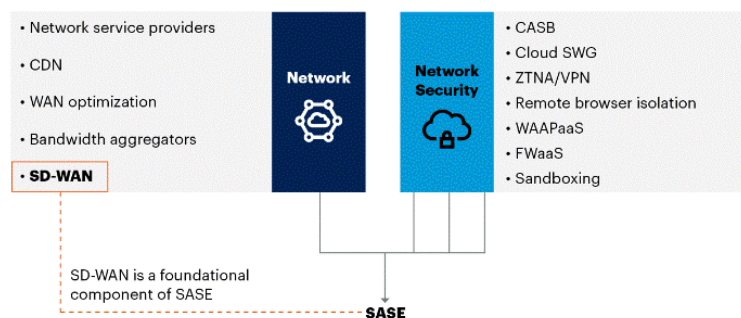


Source: Deloitte analysis of data provided by Global mobile Suppliers Association (GSA)

Security

Secure Access Service Edge (SASE) is the convergence of multiple network and security capabilities such as SD-WAN, Zero Trust Network Access (ZTNA), and Cloud Access Security Brokers (CASBs). The technology directly delivers WAN and security controls to the connection source whether it be the user, device, or IoT sensor rather than a data center. SASE models promise to deliver greater flexibility, lower latency, cost effectiveness, and increased simplicity as a cloud-native cybersecurity solution.

Exhibit 11. The Relationship between SASE and SD-WAN



Source: Gartner

SD-WAN

Software-defined wide area network (SD-WAN) is a software approach to managing the WAN. As enterprises migrate to the cloud, traditional WAN architectures experience a massive influx of network traffic from bandwidth-intensive SaaS and IaaS applications, often distributed across global hybrid IT environments. Aside from poor user experience driven by network congestion and latency, networks opened to the cloud are becoming increasingly vulnerable to security threats and complicated to manage. SD-WAN is designed to address these issues by virtualizing traditional routers, simplifying network management (users can manage multiple WAN links—both broadband and cellular), while delivering greater threat protection, higher quality of service (QoS) and cost efficiency.

In 2022, the SD-WAN market was valued at \$5B, growing faster than Wi-Fi and switching with a five-year CAGR in the mid-teens. There are about a dozen core players in SD-WAN, half of which are established Wi-Fi suppliers while others include mostly security and firewall vendors; however, the lines are blurring between the two.

EXTR is a newcomer to the space, acquiring SD-WAN capabilities in 2021 through its purchase of Ipanema, a small player with less than 5% market share. We believe EXTR will focus on integrating SD-WAN with its Wi-Fi and switching products where it has strong bases before it tries to build up a security portfolio; however, EXTR can integrate and bundle in third-party security tools instead.

Exhibit 12. ExtremeCloud SD-WAN

Feature/Benefit	Description
Unified Management	ExtremeCloud IQ enables configuration, management, and troubleshooting of Wi-Fi Aps, Ethernet switches, and SD-WAN devices.
Intelligent Application Performance Control	Automates application discovery, optimize traffic, and enforce end-to-end, per-session QoS that ensures application performance goals.
Application Performance Visualization & LAN-to-WAN-to-Cloud Performance Monitoring	Enable evidence-based decisions with actionable at-a-glance, real-time comparisons of targeted vs actual performance
Secure SaaS & IaaS Breakout	Simplifies and reduces risk of digital transformation. Policy-based internet breakout controls access to SaaS applications. Maintenance-free cloud on-ramps to leading public clouds, such as Azure and AWS, speeds cloud migration.
Fabric to the Edge	Seamless extension of Extreme Fabric to the edge over low-cost internet access improves data security and application performance management.
Built-in Zone-Based Firewall	Secures and segments SD-WAN traffic.
Choice of Cloud Security Vendor	Option for enterprises to bring own cloud security vendor enables ExtremeCloud SD-WAN to direct outbound internet traffic to their gateways.

Source: Company website

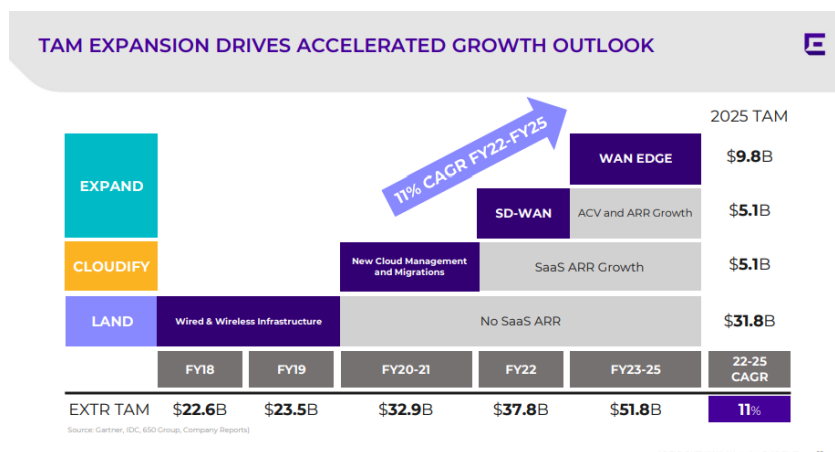
Gartner estimates the percentage of SD-WAN purchases from a single vendor SASE offering will grow from 10% in 2021 to 50% by 2025. In the same period, 10% of enterprises will use a 5G solution for a primary or backup fixed wireless connectivity in their networks, up from less than 1%, while the percentage of enterprises that automate at least half of their network activities will grow to 25%, up from 8%. On-premise NaaS adoption will rapidly accelerate from under 1% in 2021 to 15% of all enterprise by end of 2024

Industry & Competitive Overview

We have been early in predicting that virtually every industry will be radically impacted by the cloud and the use cases it is enabling, such as AI and Extended Reality (XR), alongside a variety of existing and emerging IoT applications (see our [Cloud Set to Drive a New "Roaring 20s" Defined by Big Productivity Improvements](#) report). Businesses are in the early innings of adopting cloud-based architectures that will usher in a new era of communication, data analytics, collaboration, and productivity.

EXTR has identified a \$33B total addressable market (TAM) opportunity for its portfolio of Enterprise Networking solutions, growing by approximately 11% per year for the next three years. This is comprised of a \$22B opportunity in campus networking, \$4.6B for 5G-related services, and \$2.2B for SD-WAN. EXTR also participates in a \$4B networking software market that includes network automation in addition to cloud-based and on-premises network management. The company expects this target market to grow to ~\$52B by 2025. While neither they nor we include edge compute/security in this market, we believe it could double the size of the market opportunity over time.

Exhibit 13. Extreme's Growing TAM Opportunity

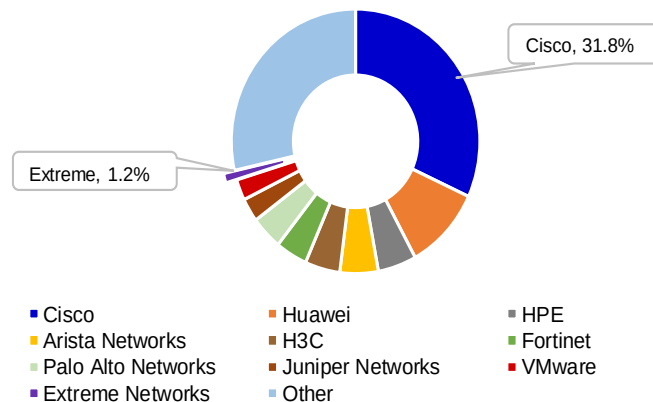


Source: Company website

As EXTR has expanded its services provided, it has moved into markets that are also faster growing. The “Land” bucket includes its basic infrastructure services, both wired and wireless. In fiscal 2020, the company began its “Cloudify” initiative with its new ExtremeCloud IQ platform and suite of management tools. Most recently, EXTR has begun to enter nascent SD-WAN and WAN edge markets, through its acquisition of Ipanema.

EXTR’s breadth and depth of end-to-end networking tools and equipment enables it to serve a gamut of organizations within a variety of different industry verticals. EXTR serves some of the world’s leading names in business, education, manufacturing, healthcare, retail, transportation & logistics, and entertainment across its global footprint. We believe EXTR has the ability to deepen its customer relationships and cross-sell its services to drive ARPUs. For example, during its fiscal year 2Q23 reporting period, 44 customers spent more than \$1M with EXTR, up from 37 the prior quarter. Its comprehensive fabric and cloud solutions are driving significant differentiation combined with its ability to deliver network automation, hyper segmentation, and security.

EXTR attributes about half of new logo wins to customers who switch due to reported fatigue and lack of innovation from its competitors, particularly the larger ones. EXTR is able to capitalize on these opportunities with its highly interoperable platform, universal hardware that offers a choice of cloud-based or on-premise deployments, and simpler licensing structures, all of which are easing common customer pain points. The company is mainly targeting high-quality businesses with annual revenues of at least \$100M up to \$2.5B and primarily serves mid-to-large size organizations with 250 to 5,000 employees for campus deployments and data centers with 1,000 or fewer employees.

Exhibit 14. CY 4Q22 Enterprise Equipment Vendor Revenue Share

Source: Gartner

The enterprise network equipment space is highly competitive, underscored by rapid technological innovation, and will be key for driving digital transformation across every industry and geography. Winners in this space will need to seamlessly converge hardware, network management software, security, specialized applications, and ultimately edge compute. The market is currently dominated by Cisco, which controls a 30%-plus revenue share vs. EXTR's sub-2% position, according to Gartner (Exhibit 14). Cisco focuses primarily on top tier customers, presenting significant opportunities for EXTR to capture share in the middle and low end markets. We believe EXTR is winning share due to the completeness, simplicity, and flexibility of its offerings whereas Cisco makes gains through acquisitions. Other notable players in the space include Huawei, Arista Networks, HPE, Fortinet, VMware, and Juniper Networks. Huawei, which operates with the second highest revenue share (10.2%), has been pushed out of the US market (where it did not have a significant presence to begin with), as well as out of many political allies including the UK, Australia, and France, but still maintains operations in Germany and Italy. Nonetheless, US- and European-based vendors tend to not compete head-to-head with peers in Asia.

Historically, it is difficult to raise prices in the enterprise networking industry, but significant pressure on supply chains have driven vendors to take pricing action. Cisco was first to raise prices of its enterprise switching, security, and wireless (APs, Controllers, etc.) beginning in October 2021 and more recently March and April 2022. We estimate this to be in the mid-20% range. EXTR followed with a high-teens to low-20% price increase throughout last year that applied to most of its products, but not services. EXTR should begin to see the full impact in the back half of this calendar year due to the staggered nature of the pricing moves which began in June 2022, but weighted more so in October and have been more selective.

EXTR's competitors all have many different product offerings, but we believe EXTR is more focused, which combined with its cloud-first strategy, enables the company to innovate faster. Most of the company's peers are behind its service offerings, particularly all the required features in software and are currently the only ones truly backwards compatible with software and hardware that can integrate with any legacy networking, thanks to EXTR's universal platforms.

EXTR was the first in the industry to ship an enterprise grade Wi-Fi 6E solution in July 2021, introduced the industry's first outdoor Wi-Fi 6E Access Point in August 2022, and was also first to market with universal platforms for enterprise class switching and wireless infrastructure as previously discussed.

Exhibit 15. ExtremeCloud IQ vs. Cisco Meraki vs. Juniper Mist

	ExtremeCloud IQ	Cisco Meraki	Juniper Mist
Description	Delivers management tools for Extreme's product portfolio plus seamless migration for enterprises with multivendor devices. Extreme Fabric is differentiator and reduces configuration burden and time to deployment while supporting granular segmentation of network users, devices, applications, and services.	For customers looking for cloud-managed networks, Meraki unifies management of wired and wireless products under single dashboard. Enterprises can automate and customize network through Cisco's ecosystem of open-source APIs and pre-made apps.	Suite of NetOps management with integrated automation. Juniper has focused on differentiating its Mist platform by providing additional AI algorithm support and conversational interface improvements via its Marvis Virtual Network Assistant (NVA).
Key Features	Digital Twins On-prem and cloud based	Network Based Application Recognition (NBAR) Integration with Cisco's Umbrella and Identity	Marvis Virtual Network Assistant Mist Edge
Gartner Ranking	Scores in top four across all use cases Excels in IoT device containment capabilities	Scores below average and bottom third in wired-only refresh/new build Scores in top half for management and administration capabilities	Scores in top two across all use cases, with slightly lower performance in security

Source: Gartner, company websites, Oppenheimer & Co. Inc.

We have long believed the industry will shift to a horizontally segmented stack as it matures, with hyperscalers as the foundation with CPaaS platforms, SaaS services, and cloud-native applications built on top. Below, we detail ten of the most significant trends in enterprise networking and security.

Ten Biggest Trends in Enterprise Networking

- Wi-Fi and cellular/5G will continue to coexist, one will not supersede the other**
 - Wi-Fi is a very entrenched technology, especially for campus and branch deployments.
 - IT experts are very familiar with Wi-Fi.
 - Enterprise Wi-Fi is a \$10B market with huge backlog due to supply chain issues, which will likely start to unwind in 1H23 and face tough y/y comps in 2H
- Enterprises want to own and control networks, but service providers increasingly moving to NaaS subscription models for recurring revenues and better ARPUs**
 - Underpinned by equipment vendors also making infrastructure available as subscription
 - EXTR is planning to launch new NaaS offering
- Big transition from on-premise network management to cloud**
 - EXTR appears well-positioned with a universal platform that can do both
- EXTR is making big changes and upgrades to sales channels**
 - Problem of many resellers not selling enough
 - Brought in Pete Brandt as SVP of Sales, will raise the bar

5. Enterprises top concern is network security

- a. Key issue right now is finding the right people to run and manage the network; networks will increasingly depend on AI/ML

6. Massive improvements to Wi-Fi 6 and 6E from 5 include better traffic management (especially for uplink) and the opening of the 6 GHz band with 6E

- a. 6 GHz is a cleaner band with less interference
- b. Still not many devices compatible
- c. Wi-Fi 7 will mainly focus on applying enhancements from 6/6E to 6 GHz band

7. Private cellular has existed for ten years with LTE for pure connectivity, but still in first inning

- a. Not a lot of end devices; challenges with integrating with other technologies
- b. Has seen some good traction past year in manufacturing, sea ports, automotive, and mining verticals
- c. Not seeing universal impact like Wi-Fi, but interest is strong

8. Networking and security are converging, and will converge with compute over time as will new forms of CDNs and even reverse CDNs

- a. Seen mostly in SD-WAN market with SASE
- b. Number of SD-WAN players building integrated security functionality and also partnering with security vendors

9. Supply chain bottlenecks are being alleviated, but will take another year to normalize

- a. EXTR has navigated supply shortages well with its universal platform that reduced the number of product SKUs, which means fewer components to chase

10. The industry will eventually see more convergence with licensed and unlicensed spectrum

- a. Standards bodies have more work to do to enable better integration
- b. Eventually, nearly all Wi-Fi vendors will be building integrations with cellular

Financial Model and Valuation

We believe EXTR is set to see strong annual top line revenue growth in the mid-to-high teens in the next several years with gross margins improving over 100 bps per year which will benefit from lower expedite fees and shipping costs paid to vendors as the supply chain loosens. There is a good chance that the company grows over 20% per year for a full year but this will be from a couple of years of catchup from pent-up demand. We expect EXTR to generate durable FCF, supported by relatively low CapEx intensity, which will help to pay down about \$7.5M in debt per quarter or \$30M per year.

EXTR has a strong balance sheet with only \$34M in net debt in 3Q'23, a leverage ratio under 1.0x, and over \$1B in assets. The company has a deferred subscription revenue balance of \$464M and a product backlog of over \$500M, 10x historical levels and equivalent to 2.5x quarterly revenue. The majority of the backlog, about 60%, is attributed to the latest generation of universal products

that will also pull through subscription and service revenues. The backlog has mainly been driven by supply chain constraints and chip shortages caused by the pandemic. Positively, EXTR has navigated these challenges and has taken action to reduce shortages, consistently qualifying more component suppliers and building inventory. Backlog is expected to remain elevated, but stable, during the next few quarters and will likely not normalize until fiscal year 2025 or 2026. However, during this period, EXTR should see significant improvements in book-to-bill ratio and should be able to begin recognizing more backlog as revenue.

Exhibit 16. EXTR Selected Performance

Extreme Selected Performance Data:

	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Y/Y Change
Product Revenue	191.1	198.4	187.1	206.3	223.4	241.1	21.5%
Service & Subscription revenue	89.8	87.1	91.1	91.4	94.9	91.4	4.9%
Total Revenue	280.9	285.5	278.2	297.7	318.3	332.5	16.5%
Normalized Gross Margin	58.2%	58.0%	57.0%	57.5%	58.5%	59.1%	1.1%
Adj. EBITDA	41.9	40.7	31.8	41.2	52.3	57.0	40.1%
Adj. EBITDA Margin (%)	14.9%	14.3%	11.4%	13.8%	16.4%	17.2%	1.5%
Normalized Operating Income	36.7	35.7	26.8	36.2	47.2	52.0	45.7%
Operating EPS	\$0.21	\$0.21	\$0.15	\$0.20	\$0.27	\$0.29	42.1%

Source: Company reports, FactSet, Oppenheimer & Co. Inc. estimates

Extreme Q4'23 Guidance:

Issued: 04/26/23

	Low	High	Mid	OPCO	Difference from Mid
Revenue (\$ mil)	340.0	350.0	345.0	344.5	-0.1%
Normalized Gross Margin (%)	59.0%	61.0%	60.0%	59.8%	-0.2%
Normalized Operating Margin (%)	15.5%	17.3%	16.4%	16.9%	0.5%
Normalized Net Income (\$ mil)	37.3	45.3	41.3	45.1	9.1%
Operating EPS (\$)	\$0.28	\$0.34	\$0.31	\$0.34	9.1%

Source: Company reports, Oppenheimer & Co. Inc. estimates

Long Term Guidance

Issued: 01/25/23

	Low	High	Mid
Product Revenue Growth	13%	15%	14.0%
Services Revenue Growth	5%	7%	6.0%
Subscription Revenue Growth	35%	45%	40.0%
Total Revenue Growth	13%	17%	15.0%
Normalized Gross Margin	64%	66%	65.0%
Normalized Operating Margin	18%	21%	19.5%
FCF as % of Revenue	15%	18%	16.5%
CapEx Intensity	1.0%	1.5%	1.3%

Source: Company reports, Oppenheimer & Co. Inc. estimates

Exhibit 17. EXTR Quarterly Financial Model (\$ millions, except per share data)

											Y/Y Change								
	2021 Full Year	2022 Full Year	Q1'23 Sep-22	Q2'23 Dec-22	Q3'23 Mar-23	Q4'23E Jun-23	2023E Full Year	2024E Full Year	2025E Full Year	2026E Full Year	2022	Q1'23	Q2'23	Q3'23	Q4'23E	2023E	2024E	2025E	2026E
EXTR Consolidated Model																			
Product	699.4	761.7	206.3	223.4	241.1	250.7	921.5	1,092.0	1,283.1	1,488.4	8.9%	11.4%	16.9%	21.5%	34.0%	21.0%	18.5%	17.5%	16.0%
Seq. growth			10.3%	8.3%	7.9%	4.0%					13.1%	10.8%	5.6%	4.9%	3.0%	6.0%	12.0%	14.0%	16.0%
Service and Subscription	310.0	350.6	91.4	94.9	91.4	93.8	371.6	416.1	474.4	550.3	16.3%	23.2%	13.9%	17.6%	16.1%	17.5%	13.4%	13.2%	12.2%
Seq. growth			0.3%	3.8%	-3.7%	2.7%					6.5%	0.3%	6.1%	0.3%	12.7%	4.6%	4.5%	8.7%	10.3%
Consolidated Revenues	1,009.4	1,112.3	297.7	318.3	332.5	344.5	1,293.1	1,508.1	1,757.5	2,038.7	10.2%	11.2%	13.3%	16.5%	23.8%	16.3%	16.6%	16.5%	16.0%
Seq. growth			7.0%	6.9%	4.4%	3.6%													
Product	310.0	360.6	99.8	103.6	108.9	111.6	423.8	480.5	544.0	610.2									
% of product revenue	44.2%	47.3%	47.9%	46.4%	44.0%	44.5%	46.0%	44.0%	42.4%	41.0%									
Service and Subscription	114.3	121.8	31.2	33.1	31.7	31.4	127.4	133.2	144.7	159.6									
% of service and subscription revenue	36.9%	34.7%	34.2%	34.9%	34.6%	33.5%	34.3%	32.0%	30.5%	29.0%									
Total Cost of Revenues	424.3	482.4	131.0	136.7	140.6	143.0	551.2	613.6	688.7	769.8									
% of revenue	42.0%	43.4%	44.0%	42.9%	42.3%	41.5%	42.6%	40.7%	39.2%	37.8%									
Adjustments to GAAP Gross Profit	28.9	17.9	4.6	4.6	4.5	4.5	18.164	18.0	18.0	18.0									
Normalized Gross Profit	614.0	647.8	171.3	186.3	196.4	206.0	760.0	912.5	1,066.8	1,286.9	5.5%	6.9%	13.9%	18.6%	30.0%	17.3%	20.1%	19.1%	18.4%
Normalized Gross Margin	60.8%	58.2%	57.5%	58.5%	59.1%	59.8%	58.8%	60.5%	61.8%	63.1%									
Research and development	197.0	190.6	51.0	52.6	54.8	56.8	215.3	245.8	286.5	332.3	(3.3%)	6.7%	9.4%	10.5%	26.0%	13.0%	14.2%	16.5%	16.0%
% of revenue	19.5%	17.1%	17.1%	16.5%	16.5%	16.5%	16.6%	16.3%	16.3%	16.3%									
Sales and marketing	276.8	294.5	78.4	80.5	84.0	86.1	329.0	377.0	430.6	489.3									
% of revenue	27.4%	26.5%	26.3%	25.3%	25.3%	25.0%	25.4%	25.0%	24.5%	24.0%									
General and administrative	66.2	68.7	18.5	24.1	21.7	19.6	84.0	93.5	109.0	126.4	3.8%	9.1%	34.7%	22.4%	22.0%	22.2%	11.4%	16.5%	16.0%
% of revenue	6.6%	6.2%	6.2%	7.6%	6.5%	5.7%	6.5%	6.2%	6.2%	6.2%									
Acquisition and integration costs	2.0	7.0	0.4	0.4	0.3	0.3	0.7	1.5	1.8	2.0	254.9%	(74.2%)	(100.0%)	(100.0%)	(7.1%)	(89.5%)	105.3%	16.5%	16.0%
% of revenue	0.2%	0.6%	0.1%	0.0%	0.0%	0.1%	0.1%	0.1%	0.1%	0.1%									
Restructuring and related charges	2.6	1.7	0.5	0.5	1.4	0.3	2.7	1.7	1.8	2.0	(37.4%)	72.4%	63.0%	234.9%	(55.3%)	52.4%	(37.7%)	5.9%	16.0%
% of revenue	0.3%	0.2%	0.2%	0.1%	0.4%	0.1%	0.2%	0.1%	0.1%	0.1%									
Amortization of intangibles	6.1	3.2	0.5	0.5	0.5	0.5	2.1	3.0	3.5	4.1	(47.1%)	(54.7%)	(37.3%)	(20.1%)	(19.1%)	(36.5%)	46.1%	17.1%	16.0%
% of revenue	0.6%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%									
Total Operating Expenses	550.7	565.8	149.3	158.2	162.4	163.8	633.7	722.5	833.1	956.2	2.7%	8.8%	12.4%	12.7%	14.0%	(10.2%)	14.0%	15.3%	14.8%
% of revenue	54.6%	50.9%	50.2%	49.7%	48.8%	47.6%	49.0%	47.9%	47.4%	46.9%									
Stock-Based Compensation	39.1	43.1	13.8	17.4	15.4	15.2	61.7	65.0	70.0	75.0									
% of revenue	3.9%	3.9%	4.6%	5.5%	4.6%	4.4%	4.8%	4.3%	4.0%	3.7%									
Other Adjustments to Operating Income	7.9	10.9	0.4	1.8	2.6	1.0	5.8	10.0	10.0	10.0									
Total Adjustments to GAAP Operating Income	47.0	54.0	14.2	19.2	18.0	16.2	67.6	75.0	80.0	85.0									
Normalized Operating Income	110.3	136.1	36.2	47.2	52.0	58.4	193.8	265.0	333.7	415.7	23.4%	(2.1%)	28.9%	45.7%	118.2%	42.4%	36.7%	26.0%	24.6%
Normalized Operating Margin	10.9%	12.2%	12.2%	14.8%	15.7%	16.9%	15.0%	17.6%	19.0%	20.4%									
Depreciation excluding Amortization	23.1	20.3	5.0	5.1	5.0	5.2	20.3	24.1	26.4	30.6									
% of revenue	2.3%	1.8%	1.7%	1.6%	1.5%	1.5%	1.6%	1.6%	1.5%	1.5%									
Normalized EBITDA	133.4	156.4	41.2	52.3	57.0	63.5	214.1	289.1	360.1	446.3	17.3%	(2.1%)	25.0%	40.1%	100.1%	36.9%	35.0%	24.6%	23.9%
% of revenue	13.2%	14.1%	13.8%	16.4%	17.2%	18.4%	16.6%	19.2%	20.5%	21.9%									
Interest income	0.4	0.4	0.4	0.9	0.8	0.8	2.9	8.0	12.0	15.0									
Interest expense	(22.9)	(12.8)	(3.8)	(3.9)	(3.9)	(3.6)	(15.2)	(14.0)	(11.9)	(9.8)									
Other (expense) income, net	(1.4)	0.5	0.3	0.2	(0.4)	-	0.2	-	-	-									
Normalized income before income taxes	86.4	124.2	33.1	44.5	48.5	55.6	181.7	259.0	333.9	421.0									
Provision (Benefit) for income taxes	14.2	20.8	6.0	8.0	9.7	10.6	34.2	57.0	73.4	92.6									
Tax rate	16.4%	16.7%	18.2%	18.0%	19.9%	19.0%	18.8%	22.0%	22.0%	22.0%									
Normalized Net Income	72.2	103.5	27.1	36.5	38.8	45.1	147.5	202.0	260.4	328.3	43.2%	(3.1%)	28.3%	41.7%	129.3%	42.5%	37.0%	28.9%	26.1%
Normalized Net Income Margin	7.2%	9.3%	9.1%	11.5%	11.7%	13.1%	11.4%	13.4%	14.8%	16.1%									
Earnings per share:																			
Operating EPS	\$0.57	\$0.77	\$0.20	\$0.27	\$0.29	\$0.34	\$1.11	\$1.53	\$1.99	\$2.54	37.0%	(2.9%)	27.5%	42.1%	127.7%	42.6%	38.4%	30.2%	27.4%
Average diluted shares	127.7	133.5	132.9	134.5	133.0	133.2	133.4	132.1	130.8	129.4	4.6%	(0.2%)	0.6%	1.0%	0.7%	(0.1%)	(1.0%)	(1.0%)	(1.0%)
Free Cash Flow Analysis:																			
Net income	72.2	103.5	27.1	36.5	38.8	45.1	147.5	202.0	260.4	328.3									
Plus: Depreciation	23.1	20.3	5.0	5.1	5.0	5.2	20.3	24.1	26.4	30.6									
Less: Capital expenditures	17.2	15.4	3.1	3.1	2.4	4.8	13.5	21.1	24.6	28.5	(10.1%)	(7.9%)	(3.4%)	(47.2%)	12.1%	(12.8%)	56.9%	16.5%	16.0%
% of revenue	1.7%	1.4%	1.1%	1.0%	0.7%	1.4%	1.0%	1.4%	1.4%	1.4%									
Less: Working Capital Adjustment	(72.3)	(65.0)	(17.6)	(29.0)	(4.3)	0.0	(51.0)	(15.0)	10.0	10.0									
Free Cash Flow	150.5	153.3	46.6	67.5	45.8	45.4	206.3	220.0	252.2	320.4	1.9%	26.5%	255.4%	NM	(24.1%)	33.9%	7.2%	14.6%	27.1%
% of revenue	14.9%	13.8%					15.9%	14.6%	14.3%	15.7%									
Less: Acquisitions	-	69.5	-	-	-	-	-	-	-	-									
Less: Stock Buybacks	-	45.0	-	49.8	25.0	23.0	97.8	99.9	101.5	102.9									
Actual FCF	127.4	(1.8)	46.6	17.7	20.8	22.4	107.5	120.1	150.7	217.5									
Balance Sheet Data:																			
Cash	246.9	194.5	198.3	202.5	203.0	217.9	217.9	308.0	428.7	616.2									
Total Debt	346.8	308.6	271.5	262.0	237.0	229.5	229.5	199.5	169.5	139.5									
Net Debt	99.9	114.1	73.2	59.5	34.0	11.6	11.6	(108.5)	(259.2)	(476.7)									
Annual interest expense	6.6%	3.9%	5.6%	5.9%	6.7%	6.0%	5.7%	7.0%	7.0%	7.0%									
Accounts Receivable, net	156.5	184.1	158.7	152.1	158.6	165.4	165.4	181.0	210.9	244.6	17.7%	22.5%	14.1%	(2.7%)	(10.2%)	(10.2%)	9.4%	16.9%	16.0%
% of revenue	15.5%	16.6%	13.3%	11.9%	11.9%	12.0%	12.8%	12.0%	12.0%	12.0%	49.7%	59.6%	71.5%	86.3%	54.0%	54.0%	(0.5%)	4.9%	3.1%
Inventories	32.9	49.2	51.8	63.8	70.3	75.8	75.8	75.4	79.1	81.5									
% of revenue	3.3%	4.4%	4.9%	5.0%	5.3%	5.5%	5.9%	5.0%	4.5%	4.0%									
Net PPE	55.0	49.6	48.0	45.7	45.2	45.2	45.2	45.2	45.2	45.2									
Total Assets	1,010.1	1,068.5	1,057.8	1,062.2	1,068.4	1,068.4	1,068.4	1,068.4	1,068.4	1,068.4									
Deferred Revenue	345.6	401.6	424.0	445.6															

Exhibit 18. EXTR DCF Model (\$ millions, except per share data)**EXTR DCF**

(\$ millions)	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E
Revenues	1,112	1,294	1,509	1,758	2,039	2,345	2,603	2,864	3,121	3,371	3,607	3,841	4,072
Growth		16.3%	16.6%	16.5%	16.0%	15.0%	11.0%	10.0%	9.0%	8.0%	7.0%	6.5%	6.0%
Adj. EBITDA	156	214	289	360	446	469	521	573	624	674	721	768	814
Margin	14.1%	16.6%	19.2%	20.5%	21.9%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%	20.0%
less: Depreciation & amortization	40	36	38	39	42	47	49	52	53	54	58	58	61
% of revenues	3.6%	2.8%	2.5%	2.2%	2.1%	2.0%	1.9%	1.8%	1.7%	1.6%	1.6%	1.5%	1.5%
EBIT	116	178	251	322	405	422	471	521	571	620	664	711	753
less: Cash taxes	22	34	55	71	89	93	104	115	126	136	146	156	166
Normalized cash taxes	18.5%	19.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%
Tax effected EBIT	95	145	196	251	316	329	368	407	446	484	518	554	588
plus: Depreciation & amortization	40	36	38	39	42	47	49	52	53	54	58	58	61
less: Cap-ex	15	13	21	25	29	34	39	43	47	51	54	58	61
% of revenues	1.4%	1.0%	1.4%	1.4%	1.4%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%
less: Acquisition Costs	70	-	-	-	-	-	-	-	-	-	-	-	-
Unlevered free cash flow	50	167	213	265	329	342	378	415	452	487	521	554	588
Growth		234.1%	27.5%	24.5%	24.2%	4.0%	10.5%	9.8%	8.8%	7.8%	7.0%	6.3%	6.0%
PV Unlevered free cash flow		167	213	238	267	250	248	245	240	233	225	215	205

Discount rate calculation:	
Risk free rate	4.0%
Risk premium	6.0%
Equity beta	1.3
Equity to capital	90%
Cost of equity	11.8%
Cost of debt	6.0%
Tax rate	22.0%
Debt to capital target	10%
unlevered FCF Terminal multiple	11.2x
Terminal Value	6,859
EBITDA Terminal Multiple	0.1x
Weighted average cost of capital	11.1%

Source: Company reports and Oppenheimer and Co., Inc.

DCF Scenario:	
Discount rate (calculation below)	11.1%
Long-term growth rate	2.0%
Long-term discount rate	11.1%
Sum of PV UFCF	2,367
PV of Terminal Value	2,349
Less Net debt	34
Plus PV of NOLs	-
= Total Equity value	4,682
Shares outstanding (diluted)	133
Per share value	\$35.20
Extreme Current Share Price	\$23.17
% Upside (downside)	51.9%

Exhibit 19. EXTR Debt Schedule

Extreme Networks, Inc.

EXTR 30226D106 2407052 NASDAQ Common stock

Instrument ID	Description	Balance Sheet Amt Out (USD)
2024		237.0
Other		32.9
-	Current Portion Of LTD	35.9
-	Unamortized Issuance Costs	-2.9
	LT Debt Total	234.1
	Current Portion of LTD	-35.9
	Net LT Debt Total	198.2
	ST Debt Total	35.9
	Total Debt	234.1

Data as of 13 Jun '23 Amount Outstanding in millions of USD

Source: FactSet DCS and FactSet Fixed Income

* The current total debt outstanding value is not considered a reconciled amount. Activity is determined on a transactional basis and there can be events that may affect amounts that are not disclosed until the next filing date.

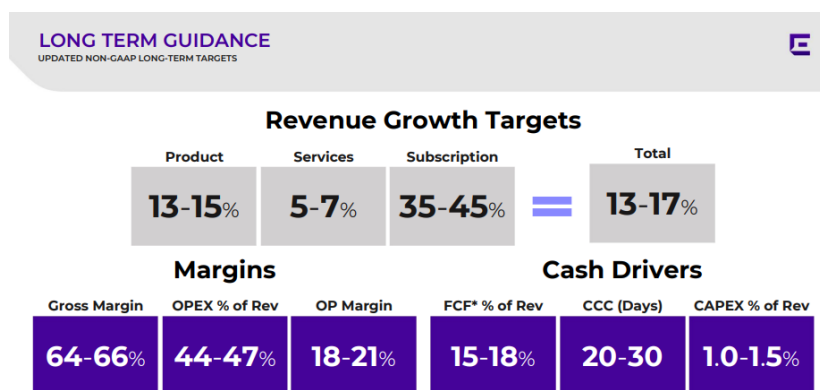
Quarterly and Long-Term Guidance

Exhibit 20. Q4'23 Non-GAAP Guidance

	FQ3'23	FQ4'23 Guidance
Revenue (\$M)	\$332.5	\$340.0 – \$350.0
Gross Margin %	59.1%	59.0% – 61.0%
Operating Margin %	15.6%	15.5% – 17.3%
EPS	\$0.29	\$0.28 – \$0.34

Source: Company website

Exhibit 21. Long-Term Guidance



Source: Company website

Note: Extreme defines FCF as Cash Flow from Operations less CapEx

Exhibit 22. Long-Term Operating Leverage Targets

STRONG OUTLOOK FOR OPERATING LEVERAGE			E
	COMMENTARY	LONG-TERM TARGET	
Cost of Goods Sold	- Supply chain costs will reduce as constraints ease - Low incremental cost to serve subscription model - Mix shift to higher margin subscription	34 – 36%	
Research & Development	- R&D investments in new products and services - Service provider support	16 – 17%	
Sales & Marketing	- Investment in Channel, APAC, Sales Enablement - Customer success to scale along with SaaS business	24 – 25%	
General & Administrative	Continue to drive efficiencies	4 – 5%	
NON-GAAP OPERATING MARGIN		18 – 21%	

Source: Company website

Management Team



Ed Meyercord is President and Chief Executive Officer. He originally joined Extreme as an independent member on the Board of Directors in 2009 before assuming his current role in 2015. Prior to his tenure at Extreme, Mr. Meyercord served as President and CEO at Cavalier Telephone & TV, a private voice, video, and data service company; in addition to Talk America, a public company that delivers phone and internet services to consumers and small businesses across the US.



Nabil Bukhari is Chief Technology Officer and Chief Product Officer. Mr. Bukhari joined Extreme as part of the Brocade acquisition and brought with him over 20 years of experience in the technology sector. Prior to joining Extreme, he held various leadership roles at companies such as Cisco, Seagate, SonicWall, and Riverstone Networks. Mr. Bukhari holds a degree in Engineering in addition to an M.B.A from London School of Economics, NYU Stern, and HEC Paris.



Kevin Rhodes, CPA is Executive Vice President and Chief Financial Officer. He most recently served as EVP and CFO at Duck Creek Technologies and has also held CFO roles at other SaaS companies including Finvi, Brighcove, and Markforged. Mr. Rhodes holds an M.B.A from Babson College's FW Olin Graduate School of Business and a B.S. in Accounting and Finance from Merrimack College.



Norman J. Rice, III is Chief Operating Officer. Mr. Rice previously served as an operating partner with Marlin Equity, a private equity firm, and has held management positions at various technology companies including Concord Communications and NitroSecurity before joining Extreme. He holds a master's degree in management and an MS in Engineering from Dartmouth College, as well as a B.S. from the University of Michigan.



Joe Vitalone is Chief Revenue Officer. He has more than 35 years of experience in sales, marketing, and operations management. Prior to Extreme, Mr. Vitalone served as EVP and President, Americas Sales and Services for Mitel and most recently was Chief Sales and Marketing Officer at Jemez Technology. He holds a B.S. in Public Relations from Western Kentucky University.



Stan Kovler is Vice President of Strategy and Investor Relations. Prior to joining Extreme, he covered companies across the Communications Equipment, Storage, and Consumer Technology sectors as the Small/Midcap IT Hardware and Networking Analyst at Citi Research. Before Citi, Mr. Kovler worked on various teams covering networking and telecom at firms such as Morgan Staley, Merrill Lynch, and Susquehanna. He holds an M.B.A from Columbia University and a bachelor's degree, with honors, from CUNY Baruch College.

Appendix

Top Comments from Recent Earnings Calls:

- Share gains evident by double-digit growth in last 8 of 9 quarters
- 60% of bookings on universal platforms for wired/wireless products
- 20% of orders from new customers (F3Q'23)
- Logo wins because of fatigue and lack of innovation from largest players in industry
 - Interoperability between platforms, simpler licensing structures
 - Universal hardware that offers choice of cloud-based and on-prem deployments
- Product lead times continue to improve as supply chain loosens, expect share gains to accelerate
- Expect backlog to remain relatively stable over next several quarters
 - Beginning stages of accelerated wave of product shipments and revenue growth over multiple quarters
 - Cancellations negligible at fraction of 1%
 - Backlog stands at \$542M, slight decrease reflects accelerated product shipments (F2Q'23)
 - Will take several quarters to come down to normalized levels (\$75-100M), probably not until F1Q'25
 - Book: bill below 1.0 by FY24
- Will see benefit in gross margin from lower expedite fees and shipping costs, but cost of components not coming down
- Pricing actions from October contributed "low- to mid-single digits" of revenue growth in December quarter
 - Think Cisco's prices are up high single-digit, low double-digit range
- Launching new products throughout 2023

Top Comments from Recent Interviews and Conferences:

- Strategy and products/services
 - Use three broad categories of edge
 - Hyperscaler
 - Telco - MEC
 - Enterprise
 - Take entire cloud environment (AWS, Azure, GCP) and run it on the enterprise edge
 - Able to do this because cloud-agnostic approach
 - Own SaaS apps
- Looking to build out 5G private solution
- Important to have solution that combines wireless LAN and WAN capabilities
 - Private 5G will be useful
 - Density and speed go Wi-Fi
 - Big campus 5G might be better
 - Have in-house ability to develop integrated partner
- Next ten years in the cloud will be very different than the past
 - Push toward edge, take cloud services and run on edge
- Enterprises will use cloud
 - SMB use service providers
- Cisco been too focused on top tier customers and partners
 - Squeezing the MSPs, presents pockets of opportunities as incoming challenger brand
- Enterprise Network Evolution
 - Pandemic revolutionized networks – first wave highly distributed, needed to connect people anywhere
 - Second wave return to campus, not return to work

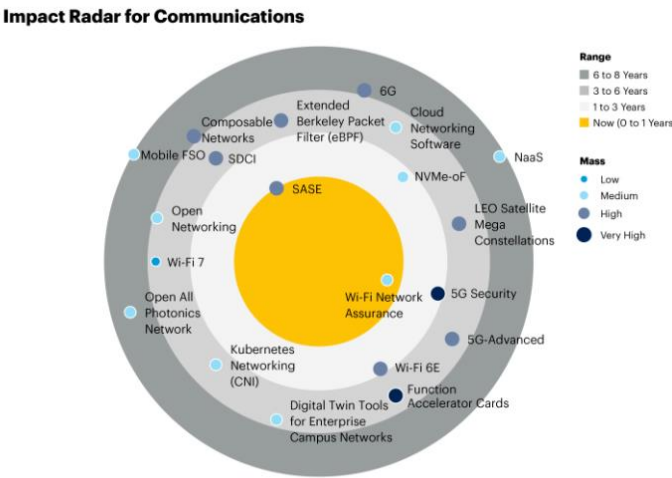
- Schools, hospitals stadiums, manufacturing, etc.
- Third wave is upgrade to Wi-Fi 6E and 5G
 - Edge switching, more AI and IoT

Exhibit 23. 2022 Gartner Magic Quadrant for Enterprise Wired and Wireless LAN Infrastructure



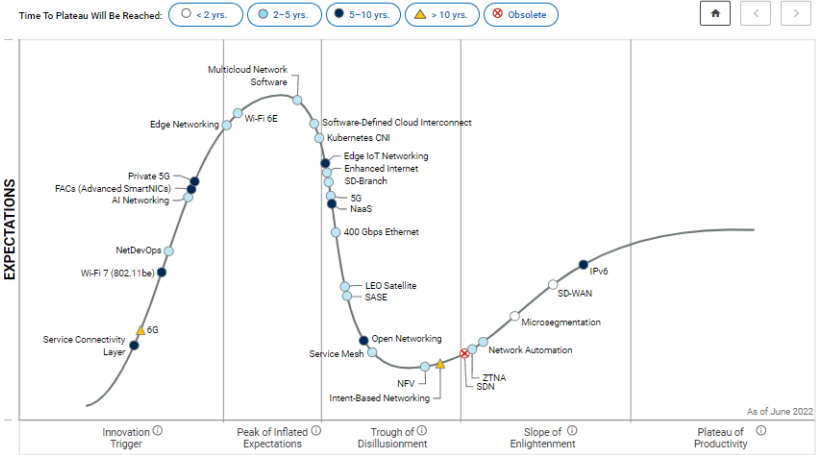
Source: Gartner (September 2022)

Exhibit 24. Gartner Impact Rader for Communications

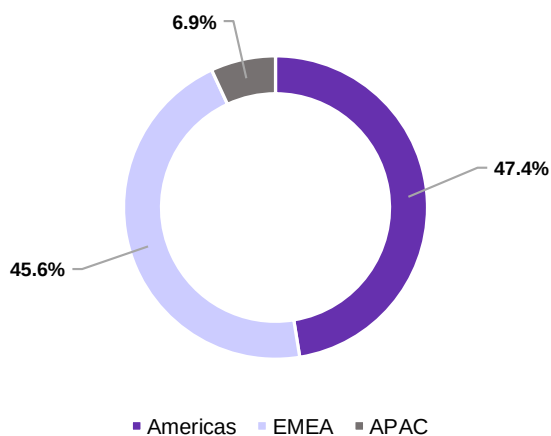


Source: Gartner

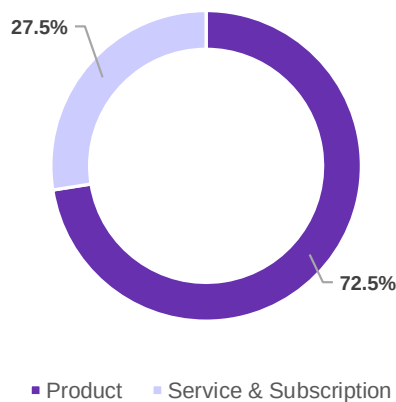
Exhibit 25. Hype Cycle for Enterprise Networking



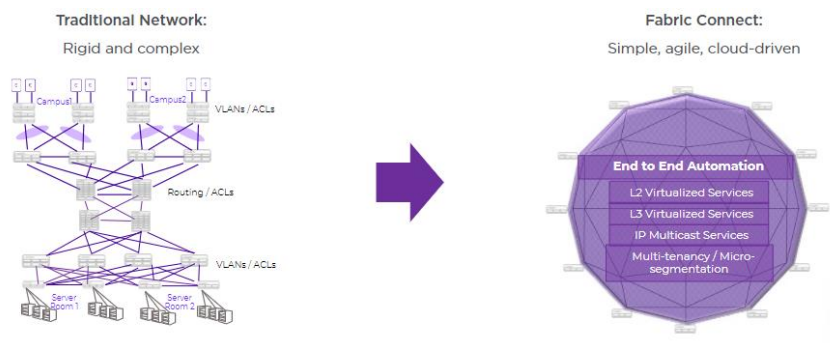
Source: Gartner (June 2022)

Exhibit 26. EXTR Q3'23 Revenue by Geography

Source: Company filings, Oppenheimer & Co. Inc.

Exhibit 27. EXTR Q3'23 Revenue by Type

Source: Company filings, Oppenheimer & Co. Inc.

Exhibit 28. Traditional Network vs. Extreme Fabric Connect

Source: Company website

Exhibit 29. Networking Definitions

Concept/Service	Description
Network Automation	Network automation tools enable the automation of network maintenance, provisioning, configuration, and reporting of network components via software. These tools interact with various hardware and software components throughout the network including switches, routers, firewalls, application delivery controllers, and CSP services.
Zero Trust Network Access (ZTNA)	ZTNA is a network security solution that grants specific users and devices access to applications based on a defined set of control policies.
Network as a Service (NaaS)	NaaS is a standardized cloud model that enables users to control, operate, and self-service the network without owning any of the network infrastructure. NaaS offerings are typically subscription or usage based. Resources can be easily and quickly scaled up or down based on real-time demand.
Network Function Virtualization (NFV)	NFV is an architecture that virtualizes network functions packaged on virtual machines running on commercial off the shelf (COTS)/commodity hardware. Traditionally, network functions such as firewalls, routers, and load balancers have ran on proprietary equipment.
Secure Access Service Edge (SASE)	SASE is the convergence of network and security service capabilities such as SD-WAN, ZTNA, and CASB. SASE models promise to deliver greater flexibility, cost effectiveness, and increased simplicity as a cloud-native cybersecurity solution.
Software Defined Wide Area Network (SD-WAN)	SD-WAN products offer greater performance, cost-effectiveness, and flexibility for enterprise WANs and support cloud architectures and modern applications. Compared to traditional routers, SD-WAN simplifies operational management and enables network traffic optimization.
Open Networking	Open Networking is a model that disaggregates hardware and software using open standards and commodity hardware. Open standards improves compatibility, interoperability, and extendibility within the network.
Edge IoT Networking	Comprised of various communications technologies that connects end-user devices and sensors to the cloud or an edge computing platform.
Private 5G	Nonpublic mobile networks that can leverage licensed, unlicensed, or shared spectrum. Private networks are designed to enhance existing network functions and unlock new capabilities not possible on public macro networks. Private enterprise networks can follow wholly owned or hybrid private-public cloud models.
Wi-Fi 7	Wi-Fi is the next generation of Wi-Fi standards (802.11be) designed for extremely high throughput. Wi-Fi 7 is still at least a year out, with ratification of the proposed standard expected in early 2024 and promises to support speeds up to 40 Gbps through more efficient use of non-contiguous spectrum.

Source: Extreme Networks, Cisco, Fortinet, VMware, Gartner, Oppenheimer & Co. Inc.

Exhibit 30. Key 5G Technologies

Technology	Description of Disruptive Wireless Technologies
Multiplexing	Allows multiple signals to be combined together and carried together. Dense Wavelength Division Multiplexing (DWDM) is an optical multiplexing technology used to increase bandwidth over existing fiber networks. DWDM works by combining and transmitting multiple signals simultaneously at different wavelengths on the same fiber.
Quadrature Amplitude Modulation (QAM)	Quadrature amplitude modulation (QAM) is a method of transmission allowing two carrier waves to be added together by modulating amplitudes of the two waves. 256-QAM is a higher order modulation (varies properties of a waveform to increase throughput) that increase throughputs of wireless spectrum by 33% versus the previous generation of 64-QAM.
Dynamic Spectrum Sharing	Enables carriers to establish a 5G coverage layer, on top of 4G spectrum without requiring spectrum re-farming.
Multiple-Input, Multiple-Output (MIMO)	This scheme enables multiple data signals over the same spectrum. It uses multiple radio antennas to multiply the capacity of radio links, exploiting special degree of freedoms. Refers to a practical technique for sending and receiving more than one data signal simultaneously over the same radio channel by exploiting multipath propagation.
Beamforming	This enables bands of spectrum that may or may not be adjacent to one another to be combined and to work in parallel. Beamforming enhances the performance of each data link, focusing a data single to eliminate interference.
Carrier Aggregation	Technique used in wireless communication to increase the data rate per user, whereby multiple frequency blocks are assigned to the same user.
C-RAN	Cloud RAN differs from Centralized RAN. The former refers to the virtualization of the entire network, allowing a server to manage between 30 and 40 cell sites per server versus 2 or 3 cell sites for each baseband unit in the case of Centralized RAN.
Software-Defined Networks (SDN)	A network architecture approach that facilitates network management by enabling a network to be centrally programmed and updated using software applications. SDN replaces traditional, proprietary network hardware SD-WAN is an important use-case.
Network Function Virtualization (NFV)	A network architecture that separately packages functions from dedicated hardware into individual network functions, which are then run on a virtual a virtual server. NFV is designed to consolidate and deliver the network components needed to support a fully virtualized infrastructure.
C-RAN	Centralized radio access networks that separate the radio and antenna from a digital baseband path. C-RAN pools multiple baseband units in a central office.
Open-RAN	It is an intelligent, software-enable RAN built on general purpose platforms with open interfaces. Lowers need for end-to-end proprietary network architecture.
Beamforming	Technology that controls the direction of a radio wave by weighing the magnitude and phase of individual antenna signals in an array of multiple antennas.
Massive MIMO	It increases capacity and system throughput with special multiplexing and time-division duplexing (TDD).
Network Slicing	A form of virtualization that allows network operators to divide RAN into multiple end-to-end networks.
Dynamic Spectrum Sharing	It is a software-based function allowing spectrum to be allocated for either 4G or 5G depending on demand. Verizon has made DSS a central part of its 5G strategy.
Phased-Array Antennas	Phased array antennas create a beam of radio waves that can be electronically steered to point in different directions without moving the antennas. Beam-steering and tracking built on MIMO create directional beams that focus a transmission to improve performance. With beamforming, the network can determine where a device, such as a smartphone or laptop, is located and project a stronger signal in that specific direction.

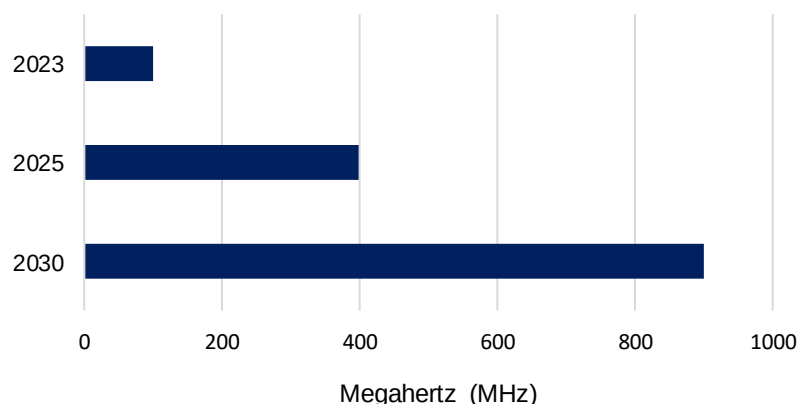
Source: SummitRidge Group "The Spectrum Handbook 2018", Gartner, Oppenheimer & Co.

Exhibit 31. Growth Opportunities in IT – Constant Dollars

	\$ Millions	2021	2022	2023	2024	2025	2026	2027	CAGR 2022-2027
Infrastructure as a Service (IaaS)		88,376	116,922	152,276	194,569	242,819	294,026	345,331	24.2%
Infrastructure Software		306,195	355,756	401,967	452,738	509,702	572,828	646,651	12.7%
Enterprise Application Software		260,321	292,476	326,497	364,490	407,003	455,086	510,075	11.8%
Vertical-Specific Software		150,191	166,202	183,426	202,336	224,523	250,125	279,330	10.9%
Consulting		229,227	256,506	281,363	307,821	335,620	368,651	405,436	9.6%
Business Process Services		185,648	201,118	214,734	229,960	245,913	263,847	284,200	7.2%
Total Sum of End-User Spending		4,300,146	4,550,151	4,788,977	5,133,246	5,461,100	5,794,135	6,167,153	6.3%
Application Implementation and Managed Services		291,377	322,705	344,632	365,404	386,236	406,739	428,920	5.9%
Mobile Phones		477,682	451,045	441,826	498,574	535,842	558,363	582,618	5.3%
Enterprise Network Equipment		56,824	66,592	72,181	75,937	78,931	81,623	84,116	4.8%
Infrastructure Implementation and Managed Services		317,845	329,339	344,422	358,664	371,841	385,579	401,312	4.0%
Servers		101,729	126,880	129,201	134,343	138,615	144,685	150,868	3.5%
Mobile Network Services		889,805	928,414	970,869	1,003,891	1,025,762	1,048,949	1,073,131	2.9%
Unified Communications		46,762	46,912	46,817	48,769	50,405	51,890	53,482	2.7%
External Controller-Based Storage		26,968	27,632	26,903	27,933	28,634	29,357	30,425	1.9%
PCs and Tablets		274,892	254,785	232,030	244,550	260,584	267,240	277,909	1.8%
Fixed Network Services		489,950	500,131	513,668	517,991	514,688	512,403	510,807	0.4%
Hardware Support		72,447	71,794	71,213	70,777	70,381	70,124	69,991	-0.5%
Printers		33,908	34,941	34,950	34,500	33,600	32,620	32,550	-1.4%

Source: Gartner

Exhibit 32. Estimated 5G Spectrum Depth – 2023E vs. 2025E vs. 2030E



Source: Gartner

Stock prices of other companies mentioned in this report (as of 06/26/2023):

Arista Networks Inc. (ANET-NYSE, \$150.82, Not Covered)

CommScope Holding Company Inc. (COMM-NASDAQ, \$4.86, Not Covered)

Fortinet Inc. (FTNT-NASDAQ, \$71.06, Not Covered)

Hewlett Packard Enterprise Co. (HPE-NYSE, \$16.39, Not Covered)

Juniper Networks Inc. (JNPR-NYSE, \$29.98, Not Covered)

Telefonaktiebolaget LM Ericsson (ERIC-NASDAQ, \$5.06, Not Covered)

Zebra Technologies Corp. (ZBRA-NASDAQ, \$273.11, Not Covered)

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Stock Prices as of June 27, 2023

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Broadcom Ltd. (AVGO - NYSE, \$821.63, OUTPERFORM)

Microsoft Corporation (MSFT - NASDAQ, \$328.60, OUTPERFORM)

Amazon.com, Inc. (AMZN - NASDAQ, \$127.33, OUTPERFORM)

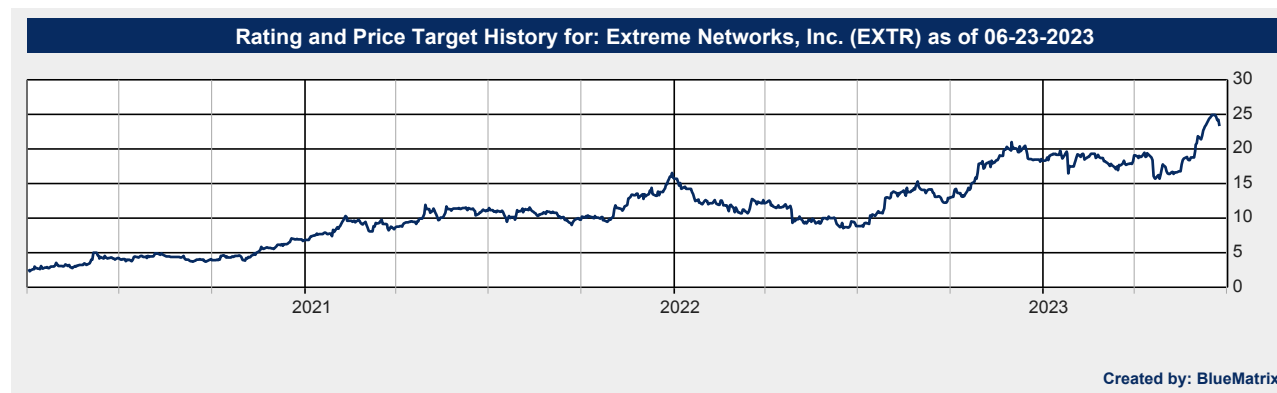
Alphabet Inc. (GOOG - NASDAQ, \$119.09, OUTPERFORM)

Comcast (CMCSA - NASDAQ, \$40.56, OUTPERFORM)

Cloudflare (NET - NYSE, \$63.19, PERFORM)

Arista Networks, Inc. (ANET - NYSE, \$150.82, PERFORM)

Fortinet, Inc. (FTNT - NASDAQ, \$71.06, OUTPERFORM)



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Distribution of Ratings/IB Services Firmwide				
Rating	IB Serv/Past 12 Mos.			
	Count	Percent	Count	Percent
OUTPERFORM [O]	459	65.29	214	46.62
PERFORM [P]	242	34.42	101	41.74
UNDERPERFORM [U]	2	0.28	1	50.00

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